

Month	2020 Sales (thousands)	2021 Sales (thousands)	2022 Sales (thousands)	2023 Sales (thousands)	Average Demand (thousands)	Season Factor	Busy or Slow
January	242	263	282	306	273	1.41	Busy
February	235	238	255	280	252	1.30	Busy
March	232	247	245	270	249	1.28	Busy
April	178	191	205	210	196	1.01	Busy
May	184	193	210	214	200	1.04	Busy
June	140	149	160	175	156	0.81	Slow
July	145	157	165	182	162	0.84	Slow
August	152	160	175	193	170	0.88	Slow
September	110	122	126	130	122	0.63	Slow
October	130	130	148	155	141	0.73	Slow
November	152	167	173	182	169	0.87	Slow
December	206	230	235	255	232	1.20	Busy

Figure 1. Season Factor

Benchmark	193.42
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Year	Period	Demand	S.F.	Deasonal	FIT	FITS	Abs Error	Error^2
2020	1	242	1.41	171	171	241	0.63	0.40
	2	235	1.30	180	172	224	11.15	124.35
	3	232	1.28	181	173	222	10.03	100.53
	4	178	1.01	176	174	176	1.95	3.80
	5	184	1.04	178	175	181	3.14	9.85
	6	140	0.81	174	176	142	1.67	2.79
	7	145	0.84	173	177	148	3.15	9.93
	8	152	0.88	173	178	156	4.07	16.58
	9	110	0.63	174	179	113	2.61	6.81
	10	130	0.73	179	179	131	0.62	0.38
	11	152	0.87	174	180	157	5.20	27.08
	12	206	1.20	172	181	217	11.13	123.86
2021	13	263	1.41	186	182	258	5.36	28.69
	14	238	1.30	183	183	239	0.86	0.74
	15	247	1.28	192	184	237	10.23	104.57
	16	191	1.01	188	185	188	3.28	10.73
	17	193	1.04	186	186	193	0.21	0.04
	18	149	0.81	185	187	151	1.96	3.85
	19	157	0.84	187	188	158	0.82	0.66
	20	160	0.88	182	189	166	6.20	38.40
	21	122	0.63	193	190	120	2.12	4.51
	22	130	0.73	179	191	139	9.00	80.98
	23	167	0.87	192	192	167	0.24	0.06
	24	230	1.20	192	193	231	0.92	0.84
2022	25	282	1.41	200	194	274	8.08	65.31
	26	255	1.30	196	195	254	1.13	1.28
	27	245	1.28	191	196	252	6.57	43.23
	28	205	1.01	202	197	199	5.60	31.38
	29	210	1.04	203	198	205	5.28	27.92
	30	160	0.81	198	199	160	0.25	0.06
	31	165	0.84	197	200	167	2.48	6.14
	32	175	0.88	199	201	176	1.32	1.75
	33	126	0.63	200	202	127	1.14	1.31
	34	148	0.73	203	203	147	0.62	0.38
	35	173	0.87	199	203	177	4.28	18.28
	36	235	1.20	196	204	245	9.71	94.20
2023	37	306	1.41	217	205	290	15.81	249.84
	38	280	1.30	215	206	269	11.12	123.73
	39	270	1.28	210	207	266	3.62	13.14
	40	210	1.01	207	208	211	1.07	1.15
	41	214	1.04	207	209	217	2.64	6.98
	42	175	0.81	217	210	170	5.46	29.76
	43	182	0.84	217	211	177	4.86	23.59
	44	193	0.88	220	212	186	6.55	42.93
	45	130	0.63	206	213	134	4.41	19.44
	46	155	0.73	213	214	156	0.76	0.59
	47	182	0.87	209	215	187	5.31	28.21
	48	255	1.20	213	216	258	3.49	12.21
2024	49		1.41		217	306		
	50		1.30		218	284		
	51		1.28		219	281		
	52		1.01		220	223		
	53		1.04		221	229		
	54		0.81		222	179		
	55		0.84		223	187		
	56		0.88		224	197		
	57		0.63		225	142		
	58		0.73		226	164		
	59		0.87		227	197		
	60		1.20		227	272		

Figure 3 Forecast Table

MAD	4.34
MSE	32.15
MAPE	2.24%

Figure 4

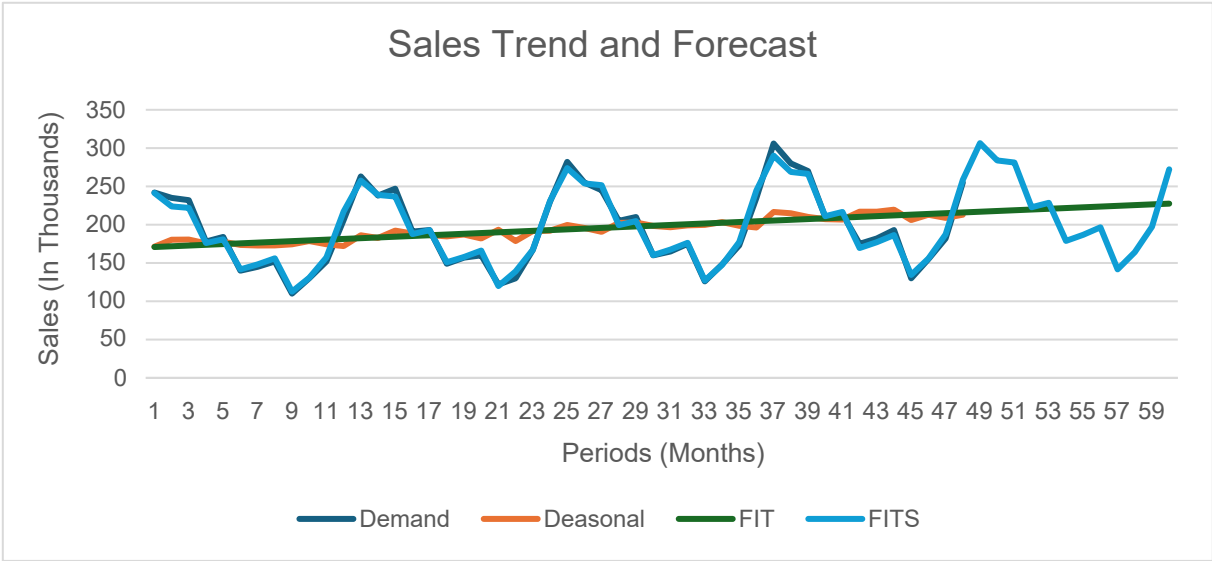


Figure 5

SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.931142424
R Square	0.867026214
Adjusted R Square	0.86413548
Standard Error	5.321438529
Observations	48

ANOVA

	df	SS	MS	F	Significance F
Regression	1	8493.410738	8493.4107	299.9328453	8.79886E-22
Residual	46	1302.614569	28.317708		
Total	47	9796.025307			

	Coefficients	Standard Error	t Stat	P-value	Lower 95%	Upper 95%	Lower 95.0%	Upper 95.0%
Intercept	169.8916368	1.560487762	108.87085	3.7654E-57	166.7505379	173.0327358	166.7505379	173.0327358
X Variable 1	0.9602053	0.055443685	17.318569	8.79886E-22	0.848602951	1.071807649	0.848602951	1.071807649

FIT = 169.89 + 0.96*t

Figure 2 Regression Model